The Three-Center Electron-Repulsion Integral Is an Elliptic Bessel Moment: Genus One at the Third Center in Momentum-Space Slater Theory*

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Two-center electron-repulsion integrals over Slater-type orbitals close in elementary functions plus the exponential integral, the logarithm, and Euler's γ—a genus-zero transcendence class—which is why the two-center problem has been solved since Roothaan and Ruedenberg. The genuine three-center integral has resisted a closed form for seventy years, and the standard verdict is that it is simply hard: reduce it to an infinite series or a one-dimensional quadrature and be done. We give it a structural reading instead. Evaluating the two-electron three-center integral $(XY|XZ)$ over $1s$ orbitals in momentum space—the Fock projection under which Coulomb–Sturmians become hyperspherical harmonics, the setting native to the Averys' generalized-Sturmian program—the three centers enter as three *phases* rather than three foci, so the coordinate obstruction (prolate spheroidal coordinates admit exactly two foci) dissolves. **[MEASURED]** The resulting momentum integral reproduces the exact integral (agreement 1.9×10^{-14} against a closed-form Gaussian reference; 4.3×10^{-7} against an independent Slater evaluation), and its angular part collapses to a single spherical Bessel kernel. **[MEASURED / classical]** What remains—the radial content—is a two-scale Bessel moment whose period is a *complete elliptic integral* (verified to 31 digits), living on the genus-one curve $y^2 = (c_1k^2 + 1)(c_2k^2 + 1)$; it degenerates to the genus-zero two-center case exactly on the diagonal $c_1 = c_2$. **[SYMBOLIC + MEASURED]** The one-mass slice is the Laplace transform of the elliptic differential—an elliptic deformation of the Bessel K_0, reducing to $K_0(D)$ when one scale decouples ($\rho \to 0$) and to the complete elliptic integral K at zero separation—and satisfies an explicit fourth-order Picard–Fuchs equation with polynomial coefficients (residual $\sim 10^{-24}$), whose characteristic exponents are the four branch points of the curve. **[OBSERVATION]** This places a molecular integral inside the modern theory of Bessel moments and two-mass elliptic Feynman integrals; a targeted literature search finds that connection unmade in either field. **[OPEN]** We do *not* give the finite closed form, and we show the natural route to it is obstructed: the elliptic-dilogarithm (unequal-mass sunrise) mechanism does not engage, because the modulus Picard–Fuchs source is not an elementary subtopology but lies in the integral's own Gauss–Manin D-module—the branch-point boundary that would supply the sunrise's tadpole vanishes identically. **[OBSERVATION]** The object is therefore an apparently unclaimed elliptic transcendent one level *above* the elliptic-dilogarithm tower, a period of a rank-four irregular connection. **[SYMBOLIC + MEASURED]** That connection is *irreducible*—it is the Fourier–Laplace transform of an irreducible connection, and its monodromy admits no invariant coordinate subspace—so no closed form arises through factorization; **[OPEN]** whether the irreducible object admits a transcendental closed form we leave open. The message is not a solution but a relocation: the third center does not make the radial integral messier within the two-center class, it raises the transcendence one genus, past the reach of the sunrise's elliptic-dilogarithm mechanism.

I. THE TWO-CENTER CLASS, AND THE WALL BEYOND IT

The electron-repulsion integral over Slater-type orbitals on two centers is one of the oldest solved problems in molecular quantum mechanics [23]. In prolate spheroidal coordinates, with the two nuclei at the foci, the Neumann expansion of $1/r_{12}$ terminates against the orbital angular momenta, and every integral reduces to elementary functions together with the exponential integral E_1, the logarithm, and Euler's constant γ. In the transcendence bookkeeping of the Geometric Vacuum program (Paper 18 [35]), these are *embedding-tier* constants—the cost of the continuum two-center geometry rather than content of the discrete graph—and, decisively, they sit at transcendence *weight one*, and no dilogarithm appears, even though the ordered radial integral has the simplex shape that generically produces one. In the direct-Coulomb $(AA|BB)$ class—and there alone—even the prefactor π cancels against the Coulomb 4π, leaving a rational combination of $\{1, e^{-\lambda R}\}$ whose zeros Lindemann–Weierstrass separates; that is what lets *that block* be *decided* rather than merely counted (Paper 58 [37]). It does not extend to the three genuinely cross classes, which carry E_1, ln and γ seeds and remain counted—weight one is not the same as decidable.

The genuine three-center integral has no such story. Prolate spheroidal coordinates are built from exactly two foci; a third nucleus has nowhere to sit. Gaussians escape this because two Gaussians on different centers multiply into a single Gaussian on a third point [22]—Slater orbitals have no product theorem—and this, not any Geometric-Vacuum-specific limitation, is the seventy-

* Paper 59 in the Geometric Vacuum series

year-old reason Gaussian bases displaced Slater orbitals for polyatomic quantum chemistry. The modern state of the art for three-center Slater repulsion integrals (Ref. [6]) reduces them to a single infinite expansion or a one-dimensional numerical quadrature; the one celebrated closed form in correlated three-electron theory, Fromm and Hill's analytic evaluation [8], is a *dilogarithm*—a genus-zero object—and belongs to a different integral topology (the one-center, explicitly correlated master integral), which lacks the structure we are about to exhibit.

This paper asks what the three-center integral *is*, transcendentally, and answers: it is an elliptic object. The two-center weight-one closure is a genus-zero fact and it stops at two centers. The third center raises the transcendence one genus, to the elliptic (genus-one) world of Bessel moments and two-mass sunrise Feynman integrals. We establish this constructively, validate every step numerically, place the object in the literature, and mark the finite closed form as open.

Lineage and scope. The momentum-space route is not new machinery. Molecular integrals over exponential-type orbitals have long been evaluated by Fourier transformation, molecular orbitals in momentum space going back to Shibuya and Wulfman [5]: the transform of a two-center orbital product is classical [24], its reduction to four-dimensional (Fock) hyperspherical harmonics goes back to Niukkanen [25], and the general one- and two-electron multicenter integrals close, via the Fourier-transform method together with Feynman's identity, to low-dimensional numerical quadrature [26]. Momentum-space Coulomb–Sturmians are hyperspherical harmonics under Fock's conformal projection [1], the framework the Averys built into the generalized-Sturmian method [2, 3, 30] and applied to Coulomb–Sturmian repulsion integrals [4]. The closest prior art to the present calculation is Avery's momentum-space route [29]—the same Fock-projection method on the same objects, and the one place the elliptic structure was nearest to visible; a targeted search finds it, like all prior work, carrying the evaluation to expansion and numerics without drawing the genus reading, though this is a search-negative inferred from the published abstracts, not a verified absence. What is new here is the reading: that the residual radial object is a specific elliptic Bessel moment, and hence that a molecular integral is an instance of the elliptic-Feynman-integral class. We work throughout with $1s$ orbitals at a common exponent; the generalization to higher (n, ℓ, m) multiplies the integrand by solid harmonics and shifts exponents but does not change the genus, and we return to this in Sec. VIII.

II. THE MOMENTUM-SPACE REDUCTION

Write the chemist integral for two densities $\rho_1 = \chi_X\chi_Y$ and $\rho_2 = \chi_X\chi_Z$, two two-center $1s$ densities that share the center X on different axes. With the Fourier representation of the kernel, $1/|\boldsymbol{r}_1 - \boldsymbol{r}_2| = (2\pi^2)^{-1}\int d^3k\, e^{i\boldsymbol{k}\cdot(\boldsymbol{r}_1-\boldsymbol{r}_2)}/k^2$, the integral becomes

$$(XY|XZ) = \frac{1}{2\pi^2}\int \frac{d^3k}{k^2}\,\tilde\rho_1(\boldsymbol{k})\,\overline{\tilde\rho_2(\boldsymbol{k})}, \qquad \tilde\rho(\boldsymbol{k}) = \int \rho(\boldsymbol{r})\, e^{i\boldsymbol{k}\cdot\boldsymbol{r}}\, d^3r. \tag{1}$$

Each center now appears only through a phase $e^{i\boldsymbol{k}\cdot\boldsymbol{R}}$: three centers are three phases, not three foci, and there is nothing to diagonalize. Equation (1) is exact; the only content is the density Fourier transform $\tilde\rho$ and the $\boldsymbol{k}$ integral.

We verify Eq. (1) two independent ways on the reference geometry $X = (0,0,0)$, $Y = (0,0,2)$, $Z = (1.5,0,0.5)$ (bohr), $\zeta = 1$, for which a McMurchie–Davidson evaluation over 8-Gaussian $1s$ fits gives the reference value $(XY|XZ) = 0.204941722$. That value carries the fit error: as the number of Gaussians grows (6 to 16) the evaluation converges to the Slater value 0.2049416525, 7×10^{-8} lower, which an exact expansion in angular channels about the shared center reproduces to 3×10^{-10} (`tests/test_pairwise_three_center.py`). **[MEASURED]** Using the closed-form Fourier transform of the Gaussian-fitted density (a product of Gaussians is a Gaussian) in Eq. (1) reproduces the McMurchie–Davidson value on the *same* fits to 1.9×10^{-14}: the momentum formula, the Fourier and Coulomb conventions, and the $\boldsymbol{k}$-quadrature are correct. Independently, computing the *true* Slater density transform (no Gaussian fit) by a one-parameter reduction of the two-center Yukawa product reproduces that reference to 4.3×10^{-7}. Both are pinned in the regression suite [39].

The angular integral closes to a spherical Bessel kernel. Writing each Slater factor through $e^{-\zeta r} = -\partial_\zeta(e^{-\zeta r}/r)$ and the Feynman parametrization of the resulting Yukawa product, the three phases combine into a *single* phase $e^{i\boldsymbol{k}\cdot\boldsymbol{W}}$ with $\boldsymbol{W}(s,t) = (t-s)\boldsymbol{X} + s\boldsymbol{Y} - t\boldsymbol{Z}$ (s,t the two Feynman parameters), so the angular integral is elementary,

$$\int d\Omega_{\boldsymbol{k}}\, e^{i\boldsymbol{k}\cdot\boldsymbol{W}} = 4\pi\, j_0(k|\boldsymbol{W}|) = 4\pi\,\frac{\sin(k|\boldsymbol{W}|)}{k|\boldsymbol{W}|}. \tag{2}$$

The whole integral reduces to a two-dimensional Feynman integral over a one-dimensional radial (k) integral,

$$(XY|XZ) = \frac{8}{\pi}\,\partial_{\zeta_a}\partial_{\zeta_b}\partial_{\zeta_c}\partial_{\zeta_d}\int_0^1 ds\int_0^1 dt\int_0^\infty dk\; j_0(k|\boldsymbol{W}|) \times \frac{e^{-D_1\Delta_1}}{\Delta_1}\,\frac{e^{-D_2\Delta_2}}{\Delta_2}\Bigg|_{\zeta=1}, \tag{3}$$

with $\Delta_1 = \sqrt{c_1k^2+1}$, $\Delta_2 = \sqrt{c_2k^2+1}$, $c_1 = s(1-s)$, $c_2 = t(1-t)$, and $D_1 = |X-Y|$, $D_2 = |X-Z|$. The ∂_ζ's (a bookkeeping device restoring the Slater from the Yukawa) and the (s,t) integrations are elementary operations; the transcendental content is entirely in the radial factor. Equation (3) reproduces the 8-Gaussian reference to 1.8×10^{-6} **[MEASURED]**, the residual being the finite-order angular and Feynman quadrature.

III. THE TRANSCENDENCE JUMP: GENUS ZERO TO GENUS ONE

The radial factor of Eq. (3) carries two dispersion factors Δ_1, Δ_2 with *different* scales $c_1 \neq c_2$. A single such factor is benign. Under Fock's substitution $k\sqrt{c} = m\sinh\theta$ it linearizes, and **[MEASURED, 6×10^{-18}]**

$$\int_0^\infty \cos(kb)\,\frac{e^{-D\sqrt{c\,k^2+m^2}}}{\sqrt{c\,k^2+m^2}}\,dk = \frac{1}{\sqrt{c}}\,K_0\Big(\tfrac{m}{\sqrt{c}}\sqrt{c\,D^2+b^2}\Big), \tag{4}$$

a Bessel K_0: the momentum-space Coulomb–Sturmian object, living on a rational (genus-zero) curve. This is why the two-center engine is weight one.

Two different scales cannot be linearized together. Their product is governed by the algebraic curve

$$y^2 = (c_1k^2+1)(c_2k^2+1), \tag{5}$$

a quartic with four distinct branch points $k = \pm i/\sqrt{c_1}, \pm i/\sqrt{c_2}$ whenever $c_1 \neq c_2$: an *elliptic curve*, genus one. When $c_1 = c_2$ the branch points collide and the quartic becomes a perfect square—rational, genus zero. The cleanest witness is the period. At $D_1 = D_2 = 0$ the radial integral is exactly a complete elliptic integral **[MEASURED, 31 digits; classical]**,

$$\int_0^\infty \frac{dk}{\sqrt{(c_1k^2+1)(c_2k^2+1)}} = \frac{1}{\sqrt{c_{\max}}}\,K(m), \tag{6}$$

with modulus $m = 1 - c_{\min}/c_{\max}$ (nondegenerate, $0 < m < 1$; prefactor verified for both orderings of c_1, c_2 in `test_paper59_euclidean_dictionary.py`), while on the diagonal $c_1 = c_2 = c$ it collapses to the elementary $\pi/(2\sqrt{c})$. Since $c_1 = s(1-s)$ and $c_2 = t(1-t)$ range independently over $[0, \frac{1}{4}]$, the full integral (3) is an integral over a *family* of elliptic curves, of modulus $m(s,t) = 1 - c_{\min}/c_{\max}$, degenerate only on the measure-zero locus $s = t$ or $s = 1-t$. Every term generated by the ∂_ζ's is a meromorphic function on the same curve (5), so the whole integral is a period of this elliptic family.

The structural statement is that the embedding tier of the transcendence taxonomy (Papers 18, 34 [35, 36]) is *genus-graded*: genus zero ($\{E_1, \ln, \gamma\}$) at two centers, genus one (elliptic) at three. The two two-center negatives—no π, no dilogarithm—are genus-zero properties; the third center does not climb to a higher rung of the polylogarithm weight tower but leaves that tower for the elliptic one. This is the transcendence-level form of "no coordinate system for three foci," and it also explains, after the fact, why an attempt to identify the three-center value against dilogarithm constants was structurally doomed: the constants are in the wrong class.

IV. THE REDUCTION: AN ELLIPTIC BESSEL MOMENT

The radial factor can be named precisely. Applying Eq. (4) to each factor and Parseval's theorem, the radial core (at fixed Feynman parameters, before the elementary j_0 and ∂_ζ wrapping) is a two-mass Bessel moment **[MEASURED, 30 digits]**,

$$M = \frac{2}{\pi\sqrt{c_1c_2}}\int_0^\infty db\; K_0\big(a_1\sqrt{p_1^2+b^2}\big)\,K_0\big(a_2\sqrt{p_2^2+b^2}\big), \tag{7}$$

with $a_i = 1/\sqrt{c_i}$ and $p_i = D_i\sqrt{c_i}$. This is a shifted-argument, two-scale member of the Bessel-moment family; it is not one of the single-scale moments $\int_0^\infty t^k K_0^n(t)\,dt$ catalogued in closed form by Bailey, Borwein, Broadhurst and Glasser [9].

A change of variable $x = \sqrt{c_1k^2+1}$ on the one-mass slice $N(D) \equiv M(D,0)$ puts it in the cleanest canonical form: the Laplace transform of the holomorphic elliptic differential,

$$N(D) = \frac{1}{\sqrt{c_1}}\int_1^\infty \frac{e^{-Dx}\,dx}{\sqrt{(x^2-1)(\rho\,x^2+1-\rho)}}, \qquad \rho = \frac{c_2}{c_1}. \tag{8}$$

This is a single explicit one-dimensional integral of an elementary integrand. A low-dimensional reduction is the expected output of the Fourier-transform method—the same machinery reduces multicenter Slater integrals to two- or three-dimensional quadrature [26] and three-center integrals to a semi-infinite oscillatory Bessel integral [27]—so what is new is not the reduction itself but the *elliptic reading* of the integrand it produces, which makes the genus-one structure manifest. It is an *elliptic deformation of the Bessel* K_0: as $\rho \to 0$ (the one-mass limit, one scale decoupling) $N \to K_0(D)$, and at $D = 0$ it is the complete elliptic integral K of Eq. (6).

Because $1/\sqrt{Q}$, $Q(x) = (x^2-1)(\rho x^2+1-\rho)$, satisfies the first-order relation $Qf' + \frac{1}{2}Q'f = 0$, and the boundary at the branch point $x = 1$ vanishes ($Q(1)^{1/2} = 0$), integration by parts against e^{-Dx} (using $\int_1^\infty x^n e^{-Dx} f\,dx = (-1)^n L^{(n)}(D)$, $L \equiv \sqrt{c_1}\,N$) yields an exact fourth-order Picard–Fuchs equation with polynomial coefficients **[SYMBOLIC PROOF; MEASURED residual $\sim 10^{-24}$]**,

$$D\rho\,L'''' + 2\rho\,L''' + D(1-2\rho)L'' + (1-2\rho)L' - D(1-\rho)L = 0. \tag{9}$$

Its characteristic polynomial factors as $\rho(s^2-1)\big(s^2 + (1-\rho)/\rho\big)$, whose roots $s = \pm 1$ and $s = \pm i\sqrt{(1-\rho)/\rho}$ are exactly the four branch points of the curve (5): the K_0/I_0 sector from the (x^2-1) pair, the oscillatory J_0/Y_0 sector from the other. (A differential-equation method for molecular two-electron integrals is itself established: Pachucki [28] evaluates the *two*-center integral from a second-order equation in the internuclear distance, closing it in elementary functions; the genus jump appears here as the step from that second-order,

genus-zero equation to the fourth-order Picard–Fuchs equation of an elliptic curve.) A subordination identity casts the same object as a two-scale sunrise integral, $\propto \iint ds_1 ds_2\,(s_1 s_2)^{-1/2} e^{-D_1^2/4s_1 - s_1 - D_2^2/4s_2 - s_2}/\sqrt{s_1 c_1 + s_2 c_2}$, whose second-Symanzik-type $\sqrt{\text{linear form}}$ is the genus-one signature and separates only when $c_1 = c_2$.

One numerical fact pins the object's class beyond the period. $N(D)$ is *non-analytic in D at $D = 0$*: it carries a $D \ln D$ term **[MEASURED]**, so it is not a pure elliptic integral (K/E are analytic in their parameters). The elliptic $D = 0$ period together with the $D \ln D$ non-analyticity are the two fingerprints of an *elliptic ⊕ logarithmic* object—the inhomogeneous ("regulator") term of the Picard–Fuchs solution associated with the elliptic dilogarithm [11, 12, 31]—a correspondence whose reach we test, and bound, in Sec. V.

[SYMBOLIC + MEASURED] *The kinematics.* Equation (4) is the frequency representation of the two-dimensional Euclidean propagator: with $G_a(\tau, x) = (2\pi)^{-1} K_0(a\sqrt{\tau^2 + x^2})$ the Green's function of $-\Delta + a^2$ on $\mathbb{R}^2$, its left-hand side is exactly $(2\pi/\sqrt{c})\, G_a(p, b)$ with mass $a = \zeta/\sqrt{c}$, Euclidean time $p = D\sqrt{c}$ and space b, and the fibre (7) is exactly the two-propagator correlator $M = 4\pi a_1 a_2 \int_{\mathbb{R}} db\, G_{a_1}(p_1, b)\, G_{a_2}(p_2, b)$: two Euclidean packets of unequal mass, evolved from a point source for Euclidean times p_1, p_2 and overlapped on one spatial slice **[MEASURED, 10^{-30} and 10^{-31} at five parameter points each;** `test_paper59_euclidean_dictionary.py`**]**. Three things become visible. First, the Feynman parameters are Källén–Lehmann spectral variables: $c = s(1 - s)$ forces $a = \zeta/\sqrt{c} \geq 2\zeta$, so each two-centre density is dispersed over its own two-particle cut and the elliptic modulus $\rho = c_2/c_1 = (a_1/a_2)^2$ is the squared ratio of the two spectral masses—the genus jump is a mass-splitting phenomenon, and the three cusps are equal masses ($\rho = 1$) and one mass at the heavy end of its cut ($\rho \to 0, \infty$). Second, the exponential twist is $a_i p_i = \zeta D_i$, *independent of the Feynman parameter*: the family is swept at fixed mass×time, and the physical point is one Compton wavelength of Euclidean-time displacement, the $u = D$ endpoint of the source-time superposition $e^{-D\Delta}\Delta^{-n} = \frac{1}{(n-1)!}\int_D^\infty (u - D)^{n-1} e^{-u\Delta}\, du$ that converts the Yukawa into the Slater. Third, $j_0(k|\boldsymbol{W}|)$ is a uniform smearing of the shared spatial coordinate over the composite separation, so the third centre enters *only* through space while the bond lengths set the two Euclidean-time offsets **[MEASURED, 1.8×10^{-10} for the full position-space fibre]**.

V. THE ELLIPTIC-DILOGARITHM ROUTE IS OBSTRUCTED

The unequal-mass sunrise is closed by variation of parameters: a second-order Picard–Fuchs operator in the external variable annihilates the two periods, its inhomogeneity is an elementary subtopology (a tadpole), and the particular solution—an iterated integral of the periods against that elementary source—is the elliptic dilogarithm [11, 12]. (The full unequal-mass sunrise has since been closed to all orders in ε as iterated integrals on $\overline{\mathcal{M}}_{1,3}$ by Bogner, Müller-Stach and Weinzierl [13]—a *Fuchsian* object; the point below is that $N(D)$ is its irregular Laplace dual and so lies outside that technology.) Running the same machinery on $N(D)$ is the natural attempt. It does not close, and the reason is structural.

The curve $y^2 = Q(x, \rho)$ of Eq. (5) moves with the modulus ρ, not with the Laplace variable D. Accordingly the two periods $K(\rho), K(1-\rho)$ are the homogeneous solutions of the second-order modulus operator **[MEASURED: both annihilated to 10^{-25}; $L(0, \rho) = K(1 - \rho)$ to 19 digits]**

$$\mathcal{M}_\rho = \rho(1 - \rho)\,\partial_\rho^2 + (1 - 2\rho)\,\partial_\rho - \tfrac{1}{4}, \tag{10}$$

and they are *not* solutions of the fourth-order operator in D: a constant gives $\mathcal{O}[1] = -D(1 - \rho) \neq 0$, and the single Bessel functions of both sectors, $K_0(D), I_0(D)$ and $J_0(\omega D), Y_0(\omega D)$ with $\omega = \sqrt{(1 - \rho)/\rho}$, are likewise not D-solutions **[MEASURED]** (they are coupled, which is why Eq. (9) does not factor into elementary Bessel operators). The elliptic-dilogarithm construction lives on the periods, so it must be run in the modulus ρ.

There it meets the obstruction. Acting with $\mathcal{M}_\rho$ under the integral sign is an exact total derivative **[SYMBOLIC PROOF; exact residual 0]**,

$$\mathcal{M}_\rho\big[Q^{-1/2}\big] = \frac{d}{dx}\Big[\, g(x)\, Q^{-1/2}\Big], \quad g(x) = \frac{-x(x^2 - 1)}{4(\rho x^2 + 1 - \rho)}, \quad g(1) = 0 \tag{11}$$

so for $D \neq 0$ the modulus source is $S(D, \rho) \equiv \mathcal{M}_\rho[L] = D\int_1^\infty e^{-Dx} g\, Q^{-1/2} dx$. Because g *vanishes at the branch point* $x = 1$, differentiating in the modulus produces no branch-point boundary term: the analogue of the sunrise's pinched propagator—the elementary tadpole that would be the inhomogeneity—is absent, and the source stays on the elliptic curve. Quantitatively **[MEASURED]**, S closes exactly and finitely only inside the integral's own Gauss–Manin D-module,

$$S(D, \rho) = \sum_{k=0}^{3} P_k(D, \rho)\, \partial_D^k L, \qquad \deg_D P_k = 2, \tag{12}$$

with held-out residual saturating flat under increase of the fit degree (below 10^{-25} in the regression witness, 5×10^{-44} at driver precision; numerically an exact finite identity); whereas the genuine variation-of-parameters shape—period level $\{L, \partial_D L\}$ plus an elementary Bessel subtopology $\{K_0, K_1\}(D)$, $\{J_0, Y_0\}(\omega D)$—only *approximates* it (residual 2×10^{-14} falling geometrically to 3×10^{-30} across fit degrees, never saturating). The inhomogeneity is in-module, not a simpler subtopology, and variation of parameters against the periods is therefore circular.

[**OBSERVATION**] $N(D)$ is thus one transcendence level above the sunrise—above the *fully solved* sunrise, not merely its leading elliptic dilogarithm: the all-orders unequal-mass sunrise is now closed as iterated integrals on $\overline{\mathcal{M}}_{1,3}$ [13], a period of a *Fuchsian* connection, whereas $N(D)$ is not. It is not an elliptic dilogarithm of elementary or period-level arguments; it is a period of the rank-four irregular connection defined by Eq. (9), for which the sunrise mechanism supplies no reduction to the elliptic-polylogarithm tower [32]. [**SYMBOLIC**] The irregularity is explicit at the operator level: at $D = \infty$ the operator of Eq. (9) admits four formal exponential solutions $e^{\lambda D}$ with $\lambda \in \{\pm 1, \pm i\sqrt{(1-\rho)/\rho}\}$—its leading Newton edge is a full-width slope-one segment, so it is irregular of Poincaré rank one, the four λ being exactly the branch points (the K_0/I_0 and J_0/Y_0 Bessel sectors), whereas the sunrise/Legendre period operator is regular singular at infinity ($\lambda = 0$ only). The two operators therefore sit on opposite sides of the Laplace transform, and the Fuchsian elliptic-polylogarithm machinery of [13] cannot reach $N(D)$. Whether $N(D)$ admits a closed form of any kind is a differential-Galois question—whether the fourth-order operator factors over the differential field generated by the periods $K(\rho), K(1-\rho)$—settled below, negatively, at the reducibility level (the operator does not factor); what remains open is only whether a *transcendental* closed form exists at the irreducible level.

[**MEASURED**] The same irregularity is visible in $N(D)$ itself, as a *resurgent* structure. Expanded about the Bessel point $\rho = 0$ ($N \to K_0(D)$), the one-mass slice has the explicit elementary term series

$$N(D) = \frac{1}{\sqrt{c_1}} \sum_{n\geq 0} (-1)^n \frac{\big((2n-1)!!\big)^2}{2^n\, n!}\, \rho^n\, \frac{K_n(D)}{D^n}, \quad (13)$$

each coefficient a rational multiple of an *integer-order* Bessel function—the transcendence does not escalate—but the series is *asymptotic*, diverging factorially (optimal truncation at $n \sim 1$–3, a few digits, then unbounded). Equivalently the $D \to \infty$ expansion $N(D) \sim e^{-D} \sum_k b_k D^{-(k+1/2)}$ is Gevrey-one, and its Borel transform is the algebraic function $\psi(\zeta) = [(\zeta+2)(\rho(1+\zeta)^2 + 1 - \rho)]^{-1/2}$ on the very curve (5), singular exactly at the three non-dominant branch points $\zeta \in \{-2, -1 \pm i\sqrt{(1-\rho)/\rho}\}$—the Stokes actions linking the e^{-D} sector to the e^{+D} and $e^{\pm i\sqrt{(1-\rho)/\rho}\, D}$ ones. A Borel–Padé resummation confirms these locations (at $\rho = \frac{1}{5}$: $\zeta = -2$ and $-1 \pm 2i$, recovered to $\sim 1\%$). So the residual transcendence of $N(D)$ is not exotic: it is the resurgence of an elementary Bessel series, its Stokes data pinned to the branch points of the elliptic curve.

[**MEASURED**] The resurgence, moreover, *closes with algebraic Stokes data*. The Borel transform $\psi(\zeta)$ carries only square-root singularities, so its amplitude at each is algebraic: $a_* = 1$ at $\zeta = -2$, and $a_*^2 = -\frac{1}{2} \mp \frac{i}{2}\sqrt{\rho/(1-\rho)}$ at $\zeta = -1 \pm i\sqrt{(1-\rho)/\rho}$ (matched to the closed-form residue to 10^{-37}, and confirmed to be the large-order growth constants by singularity analysis; the dominant Stokes ray switches from the real to the complex pair at $\rho = \frac{1}{4}$). These amplitudes *are* the Stokes constants connecting the four exponential sectors, so the entire trans-series of $N(D)$ over $\{e^{\pm D}, e^{\pm i\sqrt{(1-\rho)/\rho}\, D}\}$ has *algebraic* connection data. The one genuinely elliptic ingredient enters only at the boundary: at $D = 0$ the open Laplace thimbles close into the cycles of the curve and $\sqrt{c_1}\, N(0) = K(1-\rho)$ exactly (14 digits). So the closed form at the irregular level is explicit: $N(D)$ is the Borel–Laplace transform of the algebraic differential ψ, with algebraic Stokes data and an elliptic-period $K(1-\rho)$ normalization—as far as a genuinely irregular, genus-one object reduces. What remains transcendental is the elliptic period at the boundary and the standard Borel–Laplace resummation, not any new constant.

[**OBSERVATION**] Structurally this is the genus-one analogue of the classical resurgence of the modified Bessel function K_0—the archetypal rank-one irregular equation, whose large-argument series has an algebraic Borel transform and elementary Stokes data [14, 15]. The natural home for the algebraic-Stokes-versus-transcendental-period dichotomy seen here is Sabbah's theory of Stokes structures for Fourier–Laplace transforms of algebraic connections [16], in which the Stokes data is local at the irregular point (hence algebraic) while transcendental periods enter through the regular points (hence $K(1-\rho)$ at $D = 0$). The closest worked instance is the rank-two, genus-*zero* Bessel/Kloosterman connection treated via irregular Hodge theory [20]; and the only resurgence analysis of the sunrise/banana family we found operates in the modular parameter $q = e^{2\pi i\tau}$ (the $|q| \to 1$ limit), not the Laplace-dual D [17]. The present genus-one, Laplace-dual reading appears unmade; an absence from a targeted search, not a proof.

A concrete route from the solved dual. Because $N(D)$ is the Laplace transform of the sunrise elliptic differential and the unequal-mass sunrise is now known in full closed form [13], one concrete—if not obviously tractable—route to $N(D)$ is to Laplace-transform that solution directly, an operation that necessarily leaves the elliptic-polylogarithm class for the Bessel-moment/resurgence one, consistent with the irregular character just established. We have not carried this out.

The rank-four label is not vacuous: a base-field reducibility scan. That $N(D)$ is a period of a genuinely rank-four object—rather than one secretly carrying lower-rank content—is itself a reducibility statement, and it can be tested directly. Two structural facts constrain Eq. (9). First, the operator is *formally self-adjoint*, $L^* = L$ [**SYMBOLIC PROOF**], so its differential Galois group lies in the symplectic group $\mathrm{Sp}_4(\mathbb{C})$—the signature of a polarized period. Second, its indicial exponents at the regular singular point $D = 0$ are $\{0, 1, 1, 2\}$ [**SYMBOLIC PROOF**]; the *double* exponent at 1 is the local-monodromy source of the $D \ln D$ nonanalyticity noted above. A search over the base field $\mathbb{Q}(\rho)(D)$ then returns *no* factor of order one or two—hence, by self-adjointness,

none of order three—with polar locus at the true singularities $\{0,\infty\}$ **[SYMBOLIC, positive-control validated]**: there is no hyperexponential solution $e^{\lambda D}Q(D)$ for any $\lambda \in \{\pm 1, \pm i\omega, 0\}$, and the only two order-two factorizations the symbol $(s^2-1)(s^2+\omega^2)$ admits over $\mathbb{Q}(\rho)$—the two Bessel sectors—both fail. The two natural elementary structures are excluded by the exponents alone: L is neither the symmetric product of its Bessel sectors (which would carry exponents $\{0,0,0,0\}$) nor a symmetric cube. This is strong evidence that Eq. (9) is irreducible over $\mathbb{Q}(\rho)(D)$, and hence that the rank-four label is earned. The eigenring sharpens it to a single residual target. The endomorphism algebra of the solution space is trivial, $E(L)=\mathbb{C}$ **[computed at $\rho = 1/5$; robust across pole windows]**, so L is indecomposable; and the symplectic form pairs the exponential solution y_λ at $D=\infty$ with $y_{-\lambda}$. A non-degenerate invariant subspace would split $V = W \oplus W^\perp$ and enlarge the eigenring, so every invariant subspace is *isotropic*; the exponential torus $T=(\mathbb{C}^*)^2$ confines invariant subspaces to spans of the four lines $\{y_{\pm 1}, y_{\pm i\omega}\}$, whose two symplectic pairs (the Bessel sectors) are non-degenerate. A one-dimensional isotropic invariant subspace is a hyperexponential solution, already excluded. Hence L is reducible over $\mathbb{C}(D)$ *iff* one of the four "mixed" Lagrangian planes $\mathrm{span}\{y_{\pm 1}, y_{\pm i\omega}\}$ is invariant—equivalently, iff the Stokes matrices at $D=\infty$ [16] fail to couple the real exponential pair $\{\pm 1\}$ to the imaginary pair $\{\pm i\omega\}$. This is decidable, and it is settled twice over—once in principle, once explicitly.

In principle. The reduction is an instance of the Fourier–Laplace transform of D-modules, where irreducibility is automatic. Let $\mathcal{L}$ be the rank-one meromorphic connection on $\mathbb{P}^1$ whose flat sections are $Q^{-1/2}$: it is regular singular, with local monodromy -1 at each of the four branch points $x \in \{\pm 1, \pm i\omega\}$ and trivial monodromy at ∞ (there $Q^{-1/2} \sim x^{-2}$). Having rank one, $\mathcal{L}$ is irreducible, so its minimal (intermediate) extension $j_{!*}\mathcal{L}$ to $\mathbb{A}^1$ is a *simple* holonomic D-module. (Because every finite local monodromy is $-1 \neq 1$ and hence has no invariant, $j_!\mathcal{L} = j_{!*}\mathcal{L} = j_*\mathcal{L}$, so "the D-module generated by $Q^{-1/2}$" is unambiguous.) The Laplace kernel e^{-Dx} makes the radial integral the Fourier–Laplace transform $\mathrm{FL}(j_{!*}\mathcal{L})$, and FL is an exact auto-equivalence of the category of holonomic D-modules on the affine line (with its dual coordinate) [21]; an equivalence sends simple objects to simple objects, so $\mathrm{FL}(j_{!*}\mathcal{L})$ is simple. Its generic rank is the total drop of $\mathcal{L}$—one unit at each branch point, where the monodromy -1 leaves no invariant, and none at the regular point ∞, hence *four*, matching the order of Eq. (9), of which $N(D)$ is the defining (cyclic) solution. A simple holonomic D-module of generic rank four restricts to an irreducible connection over the function field, so **the operator of Eq. (9) is irreducible** over $\mathbb{C}(D)$, hence over $\mathbb{Q}(\rho)(D)$. Simplicity is a property of the connection at the generic point, so it forbids a factor of *any* order—apparent singularities and all; the monodromy computation next verifies it by hand.

Explicitly. The same conclusion is exhibited, and the mechanism made visible, by the monodromy. The four exponentials are the branch points $x \in \{\pm 1, \pm i\omega\}$ of $y^2 = Q$ in the Borel (Laplace-dual) plane, and the monodromy of L around $D=0$, computed in the exponential-line basis of Lefschetz-thimble solutions $\int e^{-Dx}\,dx/\sqrt{Q}$, is **[MEASURED; exact-integer certificate]** the integer unipotent matrix

$$M_0 = \begin{pmatrix} -1 & 2 & 2 & 2 \\ -2 & 3 & 2 & 2 \\ -2 & 2 & 3 & 2 \\ 2 & -2 & -2 & -1 \end{pmatrix},$$

the *same* for $\rho = \frac{1}{5}$ and $\rho = \frac{1}{3}$ (a topological invariant of the branch-point configuration), with a single 2×2 Jordan block—the lone $D\ln D$ log—and *no* invariant coordinate subspace. The Stokes matrices therefore *do* couple the real and imaginary exponential pairs, no mixed plane survives, and the reducibility criterion above returns *irreducible*—confirming the structural argument concretely. **[SYMBOLIC + MEASURED]** Consequently $N(D)$ is a period of an *irreducible* rank-four connection: it admits no closed form through any factorization of L, and the sole remaining route to a closed form is a genuinely transcendental one at the full rank-four level—the elliptic-polylogarithm / $\Gamma(2)$ multiple-modular-value structure developed in the companion (Paper 61 [38]). The differential-Galois *reducibility* question is thereby settled (negatively: L is irreducible); the transcendental closed form is the one problem that remains open [39].

VI. PLACEMENT IN THE LITERATURE

The object of Eqs. (7)–(9) sits inside a fully developed body of mathematics, applied so far only to problems far from chemistry. Two-mass elliptic Feynman integrals—the sunrise/banana family—are evaluated in terms of elliptic dilogarithms as inhomogeneous solutions of exactly this kind of Picard–Fuchs equation [11, 12, 18]; their periods are the complete elliptic integrals of Eq. (6), and at special (complex-multiplication) scale ratios their values reach critical L-values of modular forms [19, 20, 33, 34]. The closest published analog to Eq. (7) is a doubled-argument Bessel moment that Broadhurst evaluated as a product of complete elliptic integrals via a lattice-Green-function contour [10]: the same "two-scale moment $\to$ product of K's" mechanism we expect at rational ρ.

On the chemistry side the treatment is uniformly numerical: three-center Slater repulsion integrals are reduced to infinite series or one-dimensional quadrature [6, 7], and the Averys' momentum-space Coulomb–Sturmian route [4, 29], closest in spirit to ours, likewise carries the evaluation to expansion and numerics with no genus reading our search could find. We could find no paper drawing the bridge in either direction: no chemistry reference names an electron-repulsion integral as a Bessel moment or an elliptic Feynman in-

tegral, and no amplitudes reference applies the elliptic machinery to a molecular integral. **[OBSERVATION]** The identification—that the three-center Slater integral, via the Fock projection, is a two-scale Bessel moment on the sunrise elliptic curve—appears to be unmade. This is an absence from a targeted search, not a proof of absence; the residual risk is an obscure single source. **[OBSERVATION]** A second placement datum is arithmetic: the conductor-4 content this family carries (the $\mathbb{Z}[i]$ theta series θ_3^2 of the companion (Paper 61); $G = \beta(2) = L(2, \chi_{-4})$ in the corrected ring) is absent from the established Bessel-moment corpus, which sits at levels/conductors $3, 6, 8, 15$ [19, 20, 33]—a targeted search finds no Catalan-constant appearance in any of these—the nearest χ_{-4} appearance being the topological-string sector of the Broadhurst–Dorigoni programme [17]. The family thus occupies an apparently unclaimed arithmetic slot (conductor 4) in that literature, of a piece with the unmade bridge above. (Reconciliation with the value-level searches of the companion (Paper 61): the conductor-4 arithmetic is a property of the *family* and of the $D \to 0$ shadow; the physical integrated value itself shows no measured conductor-4 content at reachable heights.)

The modular and complex-multiplication structure of this elliptic family, its cosmic-Galois periods, and the Bessel-moment period algebra of the object—the master family, the Wronskian $W_0 D^{-2}$, and the quadratic period relations $B = \pi\,\Omega$ with Ω the canonical integer symplectic intersection form—are developed in the companion Paper 61 [38], which takes the present reduction as its starting point.

VII. EXPLICITLY-CORRELATED INTEGRALS ARE NATIVE TO THIS REPRESENTATION

The momentum reduction of Sec. II is *kernel-agnostic*. A two-electron matrix element of *any* operator $f(r_{12})$ between orbital products is, by the convolution theorem,

$$\langle \phi_a \phi_b \,|\, f(r_{12}) \,|\, \phi_c \phi_d \rangle = \frac{1}{(2\pi)^3} \int d^3k\; \widetilde{\rho}_{ac}(\boldsymbol{k})^*\, \widetilde{f}(\boldsymbol{k})\, \widetilde{\rho}_{bd}(\boldsymbol{k}), \tag{14}$$

the identical object as the electron-repulsion integral with the Coulomb kernel $\widetilde{(1/r_{12})} = 4\pi/k^2$ replaced by $\widetilde{f}$, and the angular closure to the spherical Bessel kernel of Eq. (2) applies unchanged. Explicitly-correlated (F12) methods take f to be a correlation factor; the standard choice is the Slater geminal $f = e^{-\gamma r_{12}}$, whose transform $\widetilde{f}(k) = 8\pi\gamma/(k^2+\gamma^2)^2$ is exact, elementary, and—unlike the Coulomb kernel—*regular* at $k = 0$. In this representation the geminal is therefore a native object: no expansion of $e^{-\gamma r_{12}}$ in Gaussians (the first workaround of Gaussian-basis F12) is required. **[MEASURED]** The momentum form (14) reproduces a direct position-space quadrature to 10^{-14}, and the Coulomb self-repulsion of a $1s$ density to its exact value $5a/8$ (`test_f12_kernel_swap_and_native_geminal`).

Whether the integral closes in elementary functions is governed *not* by the operator but by the *density topology*. **[SYMBOLIC]** When both transition densities are single-center the radial integrand is rational and the pair integral is genus zero: e.g. $\int_0^\infty [k^2/(k^2+\gamma^2)^2]\, j_0(kR)\, dk = (\pi/4\gamma)\, e^{-\gamma R}$ (`test_f12_geminal_pair_is_elementary`). A *two*-center transition density, however, contributes a Fock dispersion factor $\sqrt{c\,k^2+1}$ (Sec. IV), and two of them build the very curve of Eq. (5), $y^2 = (c_1 k^2+1)(c_2 k^2+1)$, whose period is a complete elliptic integral (`test_f12_multicenter_shares_eri_elliptic_genus`). A rational kernel—Coulomb $4\pi/k^2$ or geminal $8\pi\gamma/(k^2+\gamma^2)^2$ alike—is a rational function on this curve, so $\widetilde{f}\,dk/y$ is again an elliptic integral: the genus is *unchanged*, the geminal merely trading an elliptic integral of the first kind (the Coulomb period) for one of the third kind (its poles at $k^2 = -\gamma^2$). The transcendence class is thus set by the number of two-center transition densities—zero or one gives genus zero, two gives genus one—exactly as for the repulsion integral.

[OBSERVATION] The consequence is a *unification*, not an escape. Explicitly-correlated integrals are native to the Fock momentum representation and cost the same as the corresponding repulsion integrals, because here the difficulty of a two-electron integral lives entirely in the densities, never in the operator between them. In particular the elliptic obstruction of this paper—the genuine three-center, two-two-center-density integral—gates explicitly-correlated *evaluation* as well: the corresponding closed form (the kernel changes the kind of period, not the curve) that would evaluate the three-center repulsion integral would evaluate the multi-center F12 correction. What correlation buys is not a cheaper integral but a far more compact basis—a single r_{12} factor recovers the electron–electron cusp that orbital products approach only as $\ell \to \infty$—so the route trades many repulsion integrals for few correlated ones drawn from the same transcendence classes. We do not here treat the many-electron (resolution-of-identity) machinery of a full F12 method, nor a resource comparison with tuned Gaussian F12; the statement is about the two-electron integrals, whose momentum-space transcendence we have mapped.

VIII. SCOPE, AND THE OPEN FRONTIER

Scope. The demonstration is for $1s$ orbitals at a common exponent. Higher (n, ℓ, m) multiply the integrand of Eq. (3) by solid-harmonic polynomials in $\boldsymbol{k}$ and shift the exponents; these change the weights and arguments but not the genus of the curve (5), which is set by the two distinct scales. The result is a statement about the transcendence *class* of the three-center integral, established on its simplest genuine instance. It does not by itself deliver a polyatomic energy: a molecular Hamiltonian requires the full three-center tensor, and a value that is

part closed-form and part numerical is neither more accurate (the Gaussian fits are already exact to fit quality) nor decidable. What the reduction buys is an accurate, fast, geometry-native evaluator for the three-center integral and—the point of this paper—a precise diagnosis of why it is hard.

The open problem. As shown in Sec. V, the elliptic-dilogarithm route is obstructed: $N(D)$ is not an elliptic dilogarithm of elementary or period-level data but a period of the rank-four irregular connection (9). We settle the differential-Galois *reducibility*: the fourth-order operator is irreducible over $\mathbb{C}(D)$—hence over $\mathbb{Q}(\rho)(D)$—because it is the Fourier–Laplace transform of an irreducible connection, confirmed concretely by its exact-integer monodromy (Sec. V), so $N(D)$ admits no closed form through any factorization of the operator. The one remaining question is whether the irreducible connection nonetheless carries a *transcendental* closed form, which we do not settle. The companion (Paper 61) localizes it: the object is an Eisenstein / complex-multiplication Bessel-moment period—not a cusp-form L-value and not an elliptic dilogarithm—whose master family obeys the quadratic period relations $B = \pi\,\Omega$, Ω the canonical symplectic intersection form now identified in closed form (developed in the companion, Paper 61), preserved by the integral monodromy; the open piece is the explicit finite reduction of the observable itself to π and the CM-fibre Γ-values. (This tempers a literature-scout expectation that the object would sit at the tractable elliptic-dilogarithm level because it has only two propagator factors; that pattern-match from the sunrise is overridden by the in-module computation of Sec. V.) Because it continues the Averys' momentum-space Sturmian program and requires elliptic-Feynman-integral technique one level beyond the sunrise, it is the natural place for collaboration.

IX. CONCLUSION

The three-center electron-repulsion integral is not merely a harder version of the two-center one; it is an object of higher genus. Two centers close at genus zero over $\{E_1, \ln, \gamma\}$; three centers, read in momentum space where the coordinate obstruction dissolves, reduce to a two-scale Bessel moment on an elliptic curve, an elliptic deformation of the Bessel K_0 with an explicit fourth-order Picard–Fuchs equation and a period that is a complete elliptic integral. The seventy-year difficulty of polyatomic Slater integrals is, at the level of transcendence, a genus jump; and a classical molecular integral turns out to be an instance of the modern theory of elliptic Feynman integrals—a bridge that, to our knowledge, had not been drawn. The finite closed form remains open—and, we have shown, past the reach of the sunrise's elliptic-dilogarithm mechanism, whose inhomogeneity is absent here: the object is a period of a rank-four irregular connection which we prove irreducible—it is the Fourier–Laplace transform of an irreducible connection, and its monodromy admits no invariant coordinate subspace—so it does not close by any factorization. Whether it closes at the irreducible level—as an Eisenstein/CM Bessel-moment period, a finite reduction to π and the CM-fibre Γ-values (companion, Paper 61)—is the one frontier that remains, posed here as it deserves to be.

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